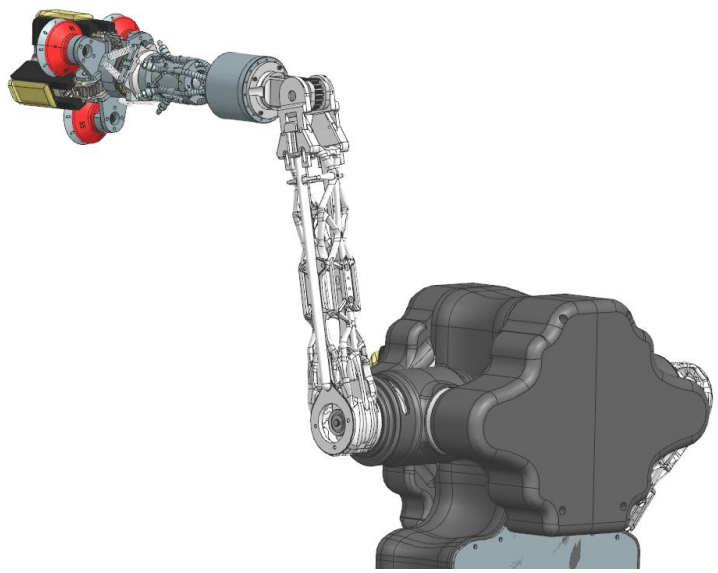
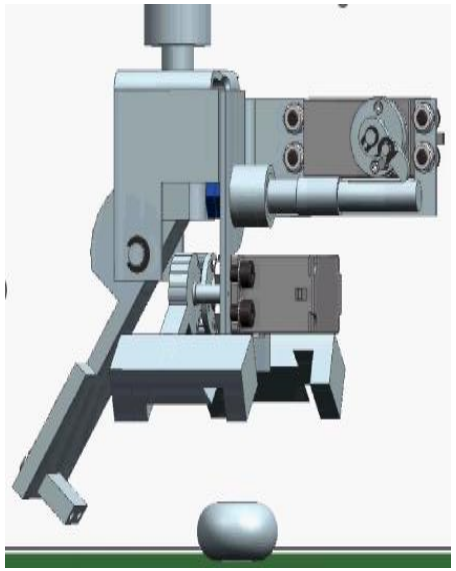
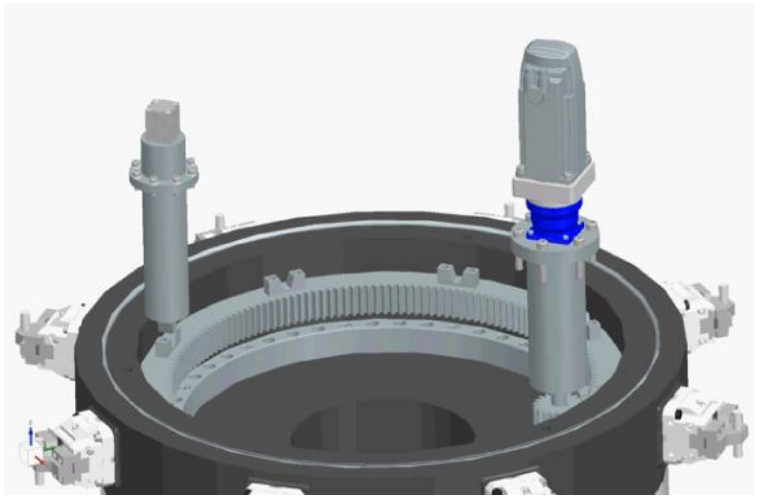
Robotics & Mechatronics Project Portfolio

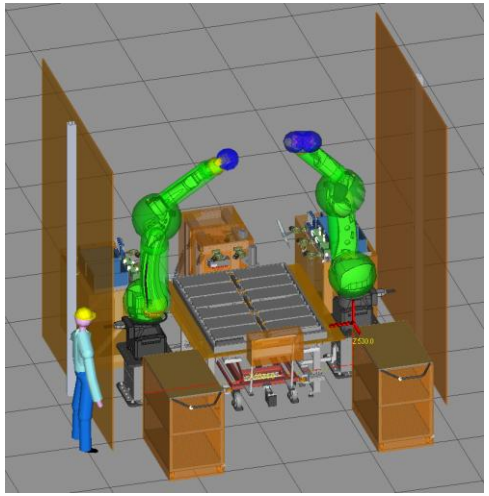
Four selected projects showing mechanical design, robotic hardware interfaces, CAD/FEM validation and prototype-oriented thinking.

Riccardo Spanò
Swiss mechanical engineering BSc graduate · SUPSI









Custom robotic hardware · mechanical validation · production-minded design · robot simulation and verification

Fail-Safe Quick-Release Joint

Robotic hardware interface designed for safety and modularity

PROBLEM

A climbing robot can be mechanically trapped even if motors and sensors still work. Losing one limb is better than losing the whole robot.

WHAT I DESIGNED

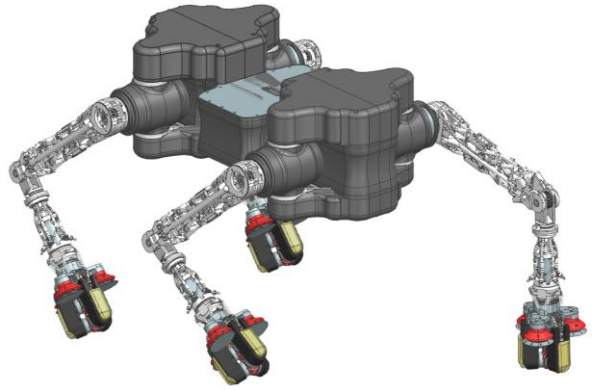
A modular energise-to-release electromagnetic joint with mechanical centering, torque transmission, electrical pogo-pin contacts and pneumatic interface.

WHY IT MATTERS

Turns a catastrophic failure mode into a controlled hardware recovery option — relevant for robots operating in debris, construction and other harsh environments.









Evidence: CAD architecture + FEM checks; <0.02 mm deformation and safety-focused fail-safe logic in the concept.

Siemens NX | FEM / Simcenter 3D | FMEA | ROS2 logic

Multi-Cobot Coordination for EV Battery Disassembly

Bachelor's thesis, SUPSI / European AutoMat project, 2026

PROBLEM

Two FANUC CR-35iB cobots shared a tool rack. Overlapping tool-change paths created interference risks.

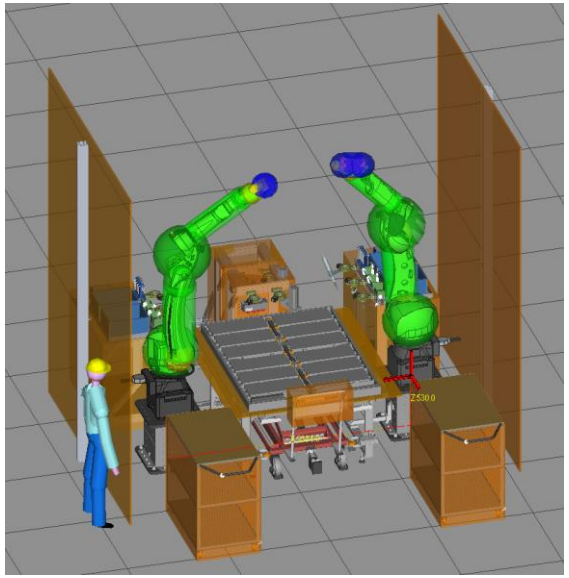
WHAT I DEVELOPED

Built IIC geometry in ROBOGUIDE, programmed tool-change sequences in FANUC TP and compared manual coordination with IIC and Deadlock Prevention.

RESULTS AND VERIFICATION

Simulation: 10 to 8 waypoints (-20%) with IIC. Preliminary cell test: automatic waiting and restart, with one robot in auto and the other in jog.

ROBOGUIDE with IIC models



Preliminary cell test



Robot programming and performance verification, with simulation results and a preliminary test on the real cell.

ROBOGUIDE | FANUC TP | IIC / DP | Test analysis

Rotary Indexing Table

Industrial indexing mechanism for stiffness, precision and manufacturability

PROBLEM

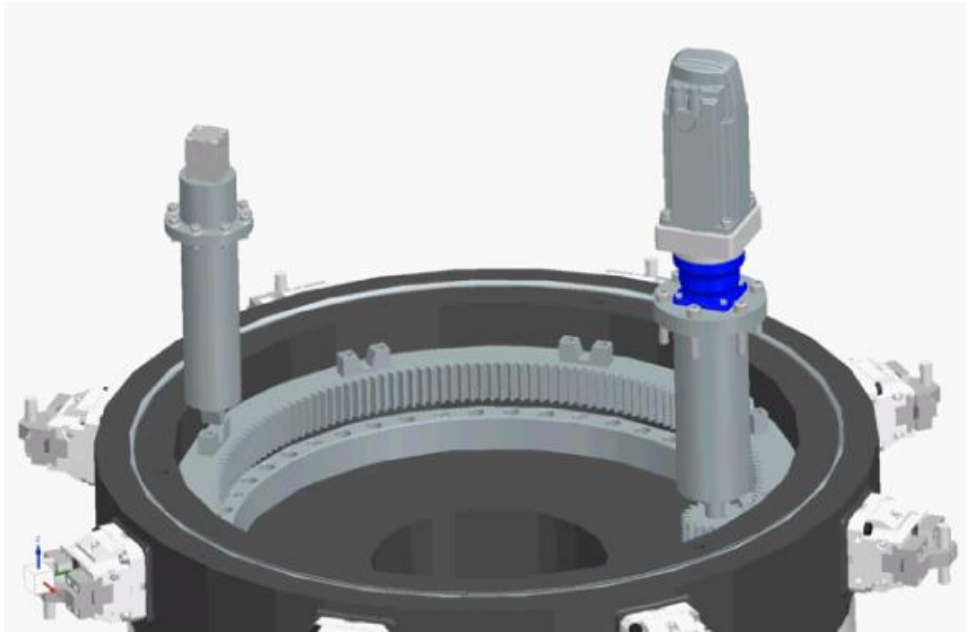
A transfer-machine table must move heavy workpieces across 8 stations while maintaining precise angular positioning under industrial loads.

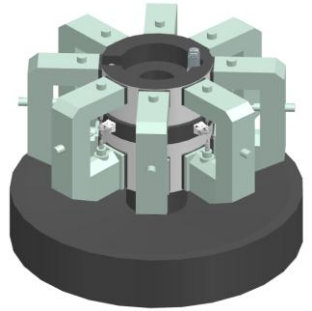
WHAT I DESIGNED

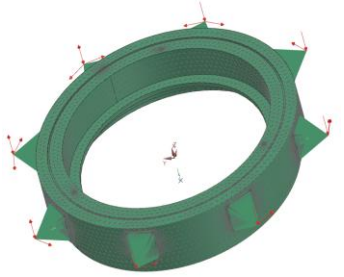
Mechanical architecture for transmission, guiding, wedge-indexing, sealing and safe lifting, with component selection and load-driven sizing.

RESULT

Concept met the stated constraints: precision within ±0.02 mm, total mass under 400 kg and stiffness above the required 50 N/µm.







Practical mechanical system design, integration of actuators/components and validation against stiffness, mass and precision targets.

3D CAD | FEM validation | Transmission | DFM mindset

Remote-Controlled Robotic Football Table

Custom XY mechanism with gripping, orientation and shooting actuation

PROBLEM

Create a compact remote-controlled table-game robot able to position a player, defend, grip/orient the disk and shoot within strict size and actuator limits.

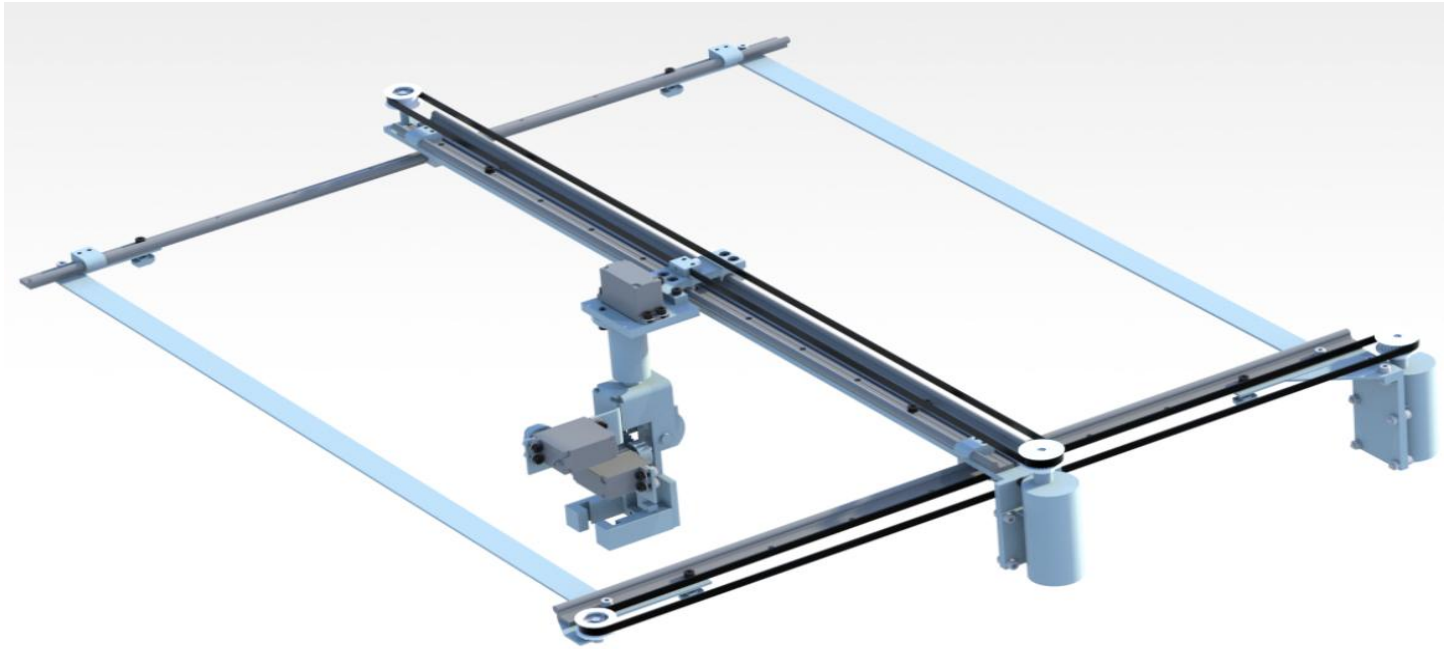
WHAT I DESIGNED

A Cartesian XY mechanism with linear guides, belt transmission, disk gripping/orientation and a spring-loaded shooting system with adjustable shot power.

WHY IT MATTERS

Shows fast custom hardware design under constraints: compact motion, actuation trade-offs, manufacturability and iterative mechanism development.

Siemens NX | Mechanisms | Belt drives | Spring sizing



Custom hardware package design with motion, actuation, compact integration and prototype-oriented mechanical thinking.